SWE 632 – User Interface Design and Development

Master of Science in Computer Science

Course Information

Course Section: SWE-632-DL1, SWE-632-004
Course Term: 2026 Spring

PROGRAM OBJECTIVES

1) **Foundations of Computer Science**: apply mathematical foundations and algorithmic principles in the modeling and design of computing systems.
2) **Computer Systems Fundamentals**: apply fundamental concepts in computer systems to model, design, and implement a computer-based system, process, or program that meets desired needs.
3) **Information Processing Fundamentals**: apply fundamental concepts in information processing to model, design, and implement a computer-based system, process, or program that meets desired needs.
4) **Advanced Computer Systems**: use advanced concepts in computer systems to design, implement, and evaluate a computer-based system, process, component, or program.
5) **Advanced Information Processing**: use advanced information processing concepts to design, implement, and evaluate a computer-based system, process, component, or program.

COURSE DESCRIPTION

Principles of user interface design, development, and programming. Includes user psychology and cognitive science, menu system design, command language design, icon and window design, graphical user interfaces, web-based user interfaces.

Recommended Prerequisite: SWE 621 or CS 531, or permission of instructor.

COURSE METHODOLOGY

Each week, learners will:

- Study content in Canvas, including video lectures, tutorials, and interactive media.
- Practice applying techniques to a real-world software application
- Work on assignments (when due)

COURSE OBJECTIVES

After completing the course, learners will be able to:

1. Design a UI through user-centered design.
2. Conduct a heuristic evaluation or think-aloud usability study to identify usability issues.
3. Use principles from visual, interaction, and community design to identify and address usability issues.
4. Improve a UI by addressing usability issues through iterative design.
5. Use modern web technologies to implement a UI.

Instructor Information

Instructor: Prof. Zhicong Lu (zlu6@gmu.edu). Office hours by appointment. Please refer to the Canvas course site for more detailed instructor contact, biography, and office hours information.

Course Resources

TEXTBOOKS AND READINGS

REQUIRED

- Norman, D. A. (2013). The Design of Everyday Things. Basic Books.
- Krug, S. (2014). Don't make me think, revisited: A common sense approach to web usability (3rd ed.). New Riders.
- Mullet, K., & Sano, D. (1994). Designing Visual Interfaces: Communication Oriented Techniques. Prentice Hall.

COMPUTER REQUIREMENTS

HARDWARE

You will need access to a Windows or Macintosh computer with at least 2 GB of RAM and to a fast, reliable broadband Internet connection (e.g., cable, DSL). The recommended computer monitor and laptop screen size is at least 13 inches. Computer speakers or headphones are recommended for recorded content. A headset microphone is recommended for live audio sessions using course tools like Canvas. Computer hard disk space must allow for:

- Installing the required and recommended software.
- Saving your course assignments.

For hardware and software purchases, visit Patriot Computers.

You are strongly encouraged to back up all contents of your computer on a regular basis. Loss of data will not excuse late or unsubmitted assignments.

SOFTWARE

Software applications include the following:

- Web browser (See Canvas Support for supported web browsers)
- Adobe Acrobat Reader (free download)
- Flash Player (free download)
- Microsoft Office (purchase)
- Zoom (free with your GMU account)

UPDATING YOUR COMPUTER

Please be sure to update your computer and prepare yourself to begin using the online format BEFORE the first day of class.

COMPUTING RESOURCES REQUIRED FOR THIS COURSE

- VPN setup: GMU Labs VPN

Grading Information

GRADING SCALE

Course letter grades will be assigned based on the following cutoffs (the minimum required percentage to achieve a specific letter grade):

GRADE	MINIMUM REQUIRED PERCENTAGE
A+	97%
A	93%
A-	90%
B+	87%
B	83%
B-	80%
C+	77%
C	73%
C-	70%
F	<70%

Late homework and project submission are NOT allowed. A submission is considered on time if submitted electronically on Canvas on or before required submission date/time.

LETTER GRADING DESCRIPTIONS:
Listed below are grades and academic standards for each grade awarded.

- **A:** Consistently performs above and beyond the course/assignment requirements
- **B:** Meets and occasionally exceeds the course/assignment requirements
- **C:** Minimally meets the course/assignment requirements
- **F:** Fails to meet the course/assignment requirements

CATEGORIES AND WEIGHTS

The following table lists the types of graded activities in this course and each category's weight in the final course grade.

ASSIGNMENT CATEGORY	% OF OVERALL COURSE GRADE
Discussions and Individual Assignments	20%
Tech Talk Presentation	5%
Group Assignments	30%
Final Project Presentation and Writeup	10%
Quizzes	10%
Final Exam	25%
Administrative	0%
TOTAL:	**100%**

All activities in this course are worth 100 points each and they are weighted by the above categories.

LATE ASSIGNMENTS
Late homework and project submission are NOT allowed. A submission is considered on time if submitted electronically on Canvas on or before required submission date/time.

UNGRADED ITEM DESCRIPTIONS (ADMINISTRATIVE)

MODULE 1 ADMINISTRATIVE - SYLLABUS AND COURSE FORMAT ACKNOWLEDGEMENT
The purpose of this activity is for you to read and understand the Syllabus for this course. Additionally, you will review the details for the course setup, so you know what to expect as you take this course.

UNGRADED DISCUSSIONS
Two non-graded discussion forums are included in this course: each with a specific purpose. *While participating in these discussions doesn't directly affect your course grade, those who regularly participate in the discussions historically perform better in the course and final exam.* Building connections with others and learning from others are keys to deeper understanding. Review each of the discussion types below.

- **Course Q and A Discussion -** Participate in this discussion to ask, view, and respond to questions of the instructor about the course material. Doing so creates a resource of information and insights. Often, even students who may not initially have a question will find a question and answer they didn't know to ask. Instructors and Tas can provide help and insights in this forum for those who may not have been able to attend the most recent office hours.
- **Cyber Café Discussion -** Participate in this discussion to discuss things that are not directly related to the course. Use this space to socialize, network, and to discuss topics of mutual interest beyond the instructional content of the course. Building personal connections, even when not directly related to course content, improves the learning experience.

GRADED ITEM DESCRIPTIONS

A submission is considered on time if submitted electronically on Canvas on or before required submission date/time. The following are descriptions of the graded categories:

DISCUSSIONS
Participate in discussions to explore topics and concepts with your peers. Discussions are an opportunity to engage with your classmates asynchronously to explore new perspectives, brainstorm ideas, and evaluate conceptual understanding. It is important to contribute to discussions in a substantive manner and follow the rules of netiquette that align with the Mason Honor Code when engaging with your peers during discussions. See each discussion for expectations on how you need to contribute to each one. You are not required to write lengthy discussion posts in this course, but you are expected to make relevant posts that further the thought line of the conversation. It is recommended that your discussion posts are "substantive" which means you addressed the questions/topics in the prompt and thoughtfully replied to a classmate on-time (or early). Posts that consist of a simple "well done" phrase or "I agree" will receive no credit.

ASSIGNMENTS
You will be required to submit homework assignments in most weeks, as specified in the course schedule. Submit all assignments via Canvas. The assignments reinforce your development of skill in applying key concepts and techniques. Some assignments are designed to be individual and will be labeled as such –individual assignments should be completed on your own. Additionally, some assignments will be labeled as 'group' and you are allowed to work with others on 'group' assignments if you choose to – if you decide to work on assignments in a group, each member of the team will receive an identical grade and all group member names must be listed on the assignment deliverable for credit to be received.

- Please note these important guidelines regarding discussion forums and group work:

- Meticulous Submission: Prior to submitting any post, it is imperative to review and ensure that the content is both relevant and free from errors. Such diligence not only enhances the quality of discussions but also showcases respect for the academic community.
- Active and Constructive Participation: All participants are encouraged to actively engage in discussions. Constructive feedback and insights are invaluable. Therefore, always aim to be supportive, concise, and articulate in your contributions.
- Subscription to Forums: For real-time updates and to stay informed of ongoing discussions, you might consider subscribing to specific forums. This will facilitate timely notifications concerning new posts, aiding in prompt engagement.
- Forum Posting Protocol: It is pertinent to note that our forum operates on a **"post-first"** system (except for the Introductions forum). This requires participants to submit their original post prior to accessing posts from peers. Such a structure is designed to foster independent thought before being influenced by others' contributions.
- Individual Assignments: some assignments in this course are designated as "Individual Assignments" and you are expected to complete these on your own.
- Working Together on Group Assignments: You will notice that some assignments in this course are designated as "Group Assignments" and are to be completed in groups ranging from one to four participants. A single team member should submit the work, ensuring the names of all other group members are mentioned in the submission to ensure they get credit.

QUIZZES

Quizzes in this course give you an opportunity to demonstrate your knowledge of the subject material. Quizzes are open-book and allow for unlimited attempts. The score on your last attempt will be used for grading.

FINAL EXAM

The Final Exam in this course gives you an opportunity to demonstrate your knowledge of the subject material. The Final Exam is closed book and only allows for one attempt.

POLICIES AND SERVICES

MASON HONOR CODE

To promote a stronger sense of mutual responsibility, respect, trust, and fairness among all members of the George Mason University community and with the desire for greater academic and personal achievement, we, the student members of the university community, have set forth this honor code:

Student members of the George Mason University community pledge not to cheat, plagiarize, steal, or lie in matters related to academic work.

You are expected to familiarize yourself with and adhere to the Honor Code. Student members of the George Mason University community pledge not to cheat, plagiarize, steal, and/or lie in matters related to academic work.

- For additional important information, including the Honor Code definitions of cheating, plagiarism, stealing, and lying, see the George Mason University Academic Integrity page.
- All students are expected to abide by the GMU Academic Standards Codes and the CS Department Honor Code. This policy is rigorously enforced. All class-related assignments are considered individual efforts unless explicitly expressed otherwise (in writing). Review the university honor code and present any questions regarding the policies to instructor. Cheating on any assignment will be prosecuted and result in a notification of the Honor Committee as outlined in the GMU Honor Code.

Three fundamental principles to follow at all times are that: (1) all work submitted be your own, as defined by the assignment; (2) when you use the work, the words, or the ideas of others, including fellow students or online sites, you give full credit through accurate citations; and (3) if you are uncertain about the ground rules on a particular assignment or exam, ask for clarification. No grade is important enough to justify academic misconduct. Use of Generative-AI tools should be used following the fundamental principles of the Honor Code. This includes being honest about the use of these tools for submitted work and including citations when using the work of others, whether individual people or Generative-AI tools.

All work performed in this course will be subject to Mason's Honor Code.

PARTICIPATION EXPECTATIONS

In this asynchronous course, active participation is both a requirement and path to your success as a student. Although we may not meet in real-time, our virtual classroom thrives on your contributions, including:

- Discussion forum posts by mid-week, with at least one follow-up reply to peers by end of week.
- Review and completion of all learning materials and interactive activities for the week.
- Timely submissions of assignments, as outlined in the course schedule.
- Student Presence: your digital footprint in the course platform is vital. Consistent engagement is monitored and contributes to your participation grade. Make your presence known by:
    - Regularly logging in and reviewing new content.
    - Actively engaging with course materials.
    - Communicating with peers and the instructor in forums.

By adhering to these guidelines, you'll not only meet the student presence requirements but also enrich our learning community with your active participation.

ACADEMIC INTEGRITY EXPECTATIONS

1. Working online requires dedication and organization. Proper preparation is expected every week. You are expected to log in to the course each week and complete the assignments and activities on or before the due dates.
2. Students must check their GMU email messages on a daily basis for course announcements, which may include reminders, revisions, and updates.
3. It is expected that you will familiarize yourself with and adhere to the Honor Code. Student members of the George Mason University community pledge not to cheat, plagiarize, steal, and/or lie in matters related to academic work.
4. It is essential that you promptly communicate any questions or problems to the instructor.

INDIVIDUALS WITH DISABILITIES

The university is committed to providing equal access to employment and educational opportunities for people with disabilities.

Mason recognizes that individuals with disabilities may need reasonable accommodations to have equally effective opportunities to participate in or benefit from the university educational programs, services, and activities, and have equal employment opportunities. The university will adhere to all applicable federal and state laws, regulations, and guidelines with respect to providing reasonable accommodations as necessary to afford equal employment opportunity and equal access to programs for qualified people with disabilities.

Applicants for admission and students requesting reasonable accommodations for a disability should call the Office of Disability Services at 703-993-2474. Employees and applicants for employment should call the Office of Equity and

Diversity Services at 703-993-8730. Questions regarding reasonable accommodations and discrimination on the basis of disability should be directed to the Americans with Disabilities Act (ADA) coordinator in the Office of Equity and Diversity Services.

EMAIL POLICY

Web: masonlive.gmu.edu

Mason uses electronic mail to provide official information to students. Examples include notices from the library, notices about academic standing, financial aid information, class materials, assignments, questions, and instructor feedback. Students are responsible for the content of university communication sent to their Mason e-mail account and are required to activate that account and check it regularly.

Students are also expected to maintain an active and accurate mailing address in order to receive communications sent through the United States Postal Service

ADDITIONAL SERVICES AND POLICIES

UNIVERSITY POLICIES

Students must follow the university policies. See University Policies.

DIVERSITY

George Mason University promotes a living and learning environment for outstanding growth and productivity among its students, faculty and staff. Through its curriculum, programs, policies, procedures, services and resources, Mason strives to maintain a quality environment for work, study and personal growth.

RESPONSIBLE USE OF COMPUTING

You are expected to adhere to the university policy for Responsible Use of Computing. See University Policies/Computing.

STUDENTS WITH DISABILITIES

Students with disabilities who seek accommodations in a course must be registered with the George Mason University Office of Disability Services (ODS) and inform their instructor, in writing, at the beginning of the semester.

UNIVERSITY LIBRARIES

University Libraries provides Library services for distance students.

WRITING CENTER

The George Mason University Writing Center staff provides a variety of resources and services (e.g., tutoring, workshops, writing guides, handbooks) intended to support students as they work to construct and share knowledge through writing.

You can now sign up for an Online Writing Lab (OWL) session just as you may sign up for a face-to-face session in the Writing Center, which means YOU set the date and time of the appointment.

COUNSELING AND PSYCHOLOGICAL SERVICES (CAPS)

The George Mason University Counseling and Psychological Services (CAPS) staff consists of professional counseling and clinical psychologists, social workers, and counselors who offer a wide range of services (e.g., individual and group counseling, workshops and outreach programs) to enhance students' personal experience and academic performance.

FAMILY EDUCATIONAL RIGHTS AND PRIVACY ACT (FERPA)

The Family Educational Rights and Privacy Act of 1974 (FERPA), also known as the "Buckley Amendment," is a federal law that gives protection to student educational records and provides students with certain rights.

Course Schedule

Please check the course website on Canvas for the detailed schedule and updates.